

Urban Wünsch

Assistant Professor

Researcher working on all things dissolved in water — disentangling the composition of complex environmental samples to understand the fate of organic substances in aquatic systems. As of late 2025: 29 publications (8 as lead), H-index 16 (Scopus), 1500+ citations.

✉ urban.wunsch@chalmers.se 📍 Gothenburg, Sweden, SE

Education

Technical University of Denmark

PhD Resolving the chemical structures responsible for the UV-visible spectroscopic properties of dissolved organic matter in aquatic environments

*Until
2018*

University of Applied Sciences Bremerhaven

M.Sc. Biotechnology

Until 2014

University of Applied Sciences Bingen

B.Sc. Biotechnology

Until 2011

Work

Chalmers University of Technology, Architecture and Civil Engineering

Assistant Professor

2025 – Present

DTU Aqua, National Institute for Aquatic Resources

Assistant Professor

2024 – 2025

Postdoc

2020 – 2024

Chalmers University of Technology, Architecture and Civil Engineering

Postdoc

2018 – 2020

Technical University of Denmark, National Institute for Aquatic Resources

PhD student

2014 – 2017

Teaching

Blue Bioeconomy Innovation

2023 – 2024

Course responsible

M.Eng, ~40 students

Ocean science & technology

2015 – 2025

Teaching topics in master-level courses

Machine learning for spectroscopic data analysis

2018 – Present

Co-organizer of online course

For Master- and PhD-students

Introduction to Chemistry & Urban spaces and functions for civil engineers

2018 –
2020

Teaching, bachelor-level courses

Large classrooms

Supervision

Frederik Lenk (B.Sc.)

2024 – 2024

Main supervisor

Special course

Maria Papadimitraki (Research Assistant)

2023 – 2023

Co-supervisor

University of Southern Denmark

Emilie Tage Andresen (B.Sc.)

2022 – 2022

Co-supervisor

DTU Chemistry (Denmark)

Linea Gry Ebbesen (B.Eng.)

2022 – 2022

Co-supervisor

DTU Environment (Denmark)

Laila Vinther & Cecilie Hjelm Hvas Hansen (M.Sc.)

2021 – 2021

Co-supervisor, voluntary special course

DTU Environment (Denmark)

Evelyn Namuga Kasule (B.Sc.)

2021 – 2021

Co-supervisor

DTU Environment (Denmark)

Signe Melbye Andersen (M.Sc.)

2020 – 2022

Co-supervisor

DTU Aqua (Denmark)

Anders Dalhoff Bruhn Jensen (PhD)

2019 – 2023

Co-supervisor

DTU Aqua (Denmark)

Xianyu Kong (PhD)

2019 – 2022

PhD committee member

Alfred Wegener Institute (Germany)

Nashita Moona (PhD)

2018 – 2021

Co-supervisor

Chalmers (Sweden)

Projects

Patent application (DTU)

2024 – Present

Determination of dissolved organic matter photosensitivity based on fluorescence ratios

Main author

drEEM toolbox for MATLAB

dreem.openfluor.org

2018 – Present

MATLAB toolbox for the multiway decomposition of fluorescence EEMs into underlying fluorescence components (PARAFAC).

Main developer

Skills

Dissolved organic matter characterization

EEM spectroscopy Absorbance spectroscopy

HPSEC FT-ICR-MS

Chemometrics

PCA

PARAFAC & PARAFAC2

N-PLS

ACMTF

A new approach to laccase discovery from the sea

2023

Independent Research Fund Denmark

2.5 million DKK, named applicant

New eyes on old GAC

2022

FORMAS, Sweden

146,000 SEK, co-applicant

Oxidation Demand and Rate (ODaR) as a sensitive proxy of biostability in drinking water

2021

FORMAS, Sweden

26,000 SEK, co-applicant

Enhancing Biogeochemical Fingerprints of Natural Organic Matter with Data Fusion

2019

Åforsk Foundation

400,000 SEK, main applicant

An optical sensor for reducing biocide dependence in chemical manufacturing

2019

Vinnova, Swedish Governmental Agency for Innovation Systems

474,000 SEK, co-applicant

2-year postdoctoral scholarship

2019

Carl Tryggers Stiftelse

552,000 SEK, main author

Invited participation in the Dissertations Symposium in Chemical Oceanography

2018

National Science Foundation & National Oceanic and Atmospheric Administration

Student travel award and invited participation

2018

International Humic Substances Society

Scholarship for conference participation

2017

Otto Mønsted Fond

Scholarship for external research stay

2015

Kaj og Hermilla Ostenfeld's Fond

Conferences

Photosensitivity of fluorescent dissolved organic matter across the arctic ocean²⁰²³

Aquatic Sciences Meeting 2023, Mallorca, Spain

Combining Physical and Mathematical Chromatography to Decipher the Molecular Fingerprints of Dissolved Organic Matter²⁰²¹

20th Conference of the International Humic Substances Society (online)

Mathematical chromatography deciphers the molecular fingerprints of dissolved organic matter²⁰²¹

Aquatic Sciences Meeting 2021 (online)

Interests

Association for the Sciences of Limnology and Oceanography (ASLO)

Member since 2014

International Humic Substances Society (IHSS)

Member since 2018